

Bunnyhopping in CS2

zer0.k

Abstract

This is a writeup of how bunnyhopping works in CS2 compared to CS:GO, and why perfect bops in CS2 feel inconsistent (possibly).

It is based on reverse engineering CS2's movement code.

The numbers were derived from the code and then checked in game with about 130,000 automated bot landings (see Measured in game).

Keywords: bunnyhopping; CS2; CS:GO; subtick; jump buffering.

Markdown version: github.com/zer0k-z/cs2-movement-issue-bhop.

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1 Summary (TL;DR)

CS:GO was simple. It was tick perfect, meaning the jump had to be in the first tick on the ground, which gave a one tick window and at best a 50% perf rate at the ideal scroll speed. CS2 works differently in several ways, and each one changes what a player should do.

- **Scroll at about 16ms per notch (50 to 60 per second).** The jump cooldown is a hard 15.625ms cutoff, and because press times are rounded, anything under about 15.9ms between presses is partly or fully locked out. It's better to be a little slow than a little fast.
 - Coming from 64 tick CS:GO, scroll about twice as fast as before.
 - Coming from 128 tick CS:GO, scroll a little slower than before. Going slightly faster than 15.625ms per scroll step used to still give you a somewhat decent chance to jump, but that is no longer the case in CS2.
 - At 16ms per jump input, a normal flat bhop is perfect about 49% of the time in game (64% counting restored bhops).
- **Set fps_max to a multiple of 62.5 or slightly below it (125, 250, 500...), and avoid the range just above it up to the next multiple of 64 (e.g. 252–256, 504–512).** Your scroll steps get rounded to the frame grid, and on those caps the frame step closest to the ideal timing is exactly 16ms, which always gets through. See fps_max and consistent bhop timing.
- **Don't press or release movement keys right as you land.** Even with sv_bhop_time_window, CS2 is still mostly frame perfect, so a true perfect bhop almost always needs the jump to be the very next input after the player is (effectively) on the ground. Any other key press or release in between (strafe keys, W, duck, attacking while turning) turns it into a restored bhop or a miss. Finish the A/D strafe switch before touchdown, or do it after the jump.
- **Perfect bhops are impossible at low speed.** A jump inside the bhop window can be a real perfect bhop (frame perfect or buffered), a restored bhop (speed kept, but the jump starts later and from the floor, so its height differs), or a deferred jump (a full tick on the ground). Landing at or below max speed always gives a deferred jump no matter the timing, so you will never get the extra height from perfect bhops (which is what matters here).
- **Bhopping up onto higher ground is easier than flat ground, and dropping down is harder.** A jump input while you're within 2 units of the floor counts as a perfect bhop, as long as you have enough horizontal speed and it's your first input in that range before a tick ends (after a tick end it can still be restored, which keeps your speed but not your height). The time you spend in those 2 units is roughly inversely proportional to how fast you're falling, so the slower you land, the bigger the window. With perfect scrolling you keep your speed about 63% of the time on flat ground, up to about 74% when landing on something 24+ units higher (it stops improving past that), and only about 48% after a big drop (1000 u/s).
- **Ledgegrabs don't really interact with the new bhop mechanics.** Ledgegrabbing (landing on the ground while still going up) completely bypasses the new bhop mechanics, because a landing is only recorded when you're falling. There is no window and no restore, and the jump only keeps your speed if it comes after you're within 2 units of the ledge and before the next tick ends or any other input happens. Depending on timing that's anywhere from nothing to one tick, instead of CS:GO's full tick. The same issue applies to crouchbugs/jumpbugs.
- **For jumpbugs, use a mouse wheel +alias/-alias bind and consider a lower fps cap.** A jumpbug needs the unduck and the jump in the same client frame, which is only practical when one scroll step sends both. Frame perfect inputs get easier at lower fps because frames are longer. On high fps values, a legitimate jumpbug with no alias could easily take 5 minutes to hit, even for an experienced player.

2 The basics

2.1 Ticks

The server simulates movement in fixed steps called ticks. CS2 runs at 64 ticks per second, so one tick is 15.625ms. CS:GO ran at 64 on matchmaking and 128 on most community and third party servers (7.8125ms per tick).

2.2 What a perfect bhop is

When the player lands, the ground movement code (WalkMove) starts running. The first time it runs, it clamps the player's speed down to their max speed (depends on the weapon), which is temporarily lowered even further right after a landing. After a normal flat bhop landing that is about 75% of max speed, so a single WalkMove takes a 275 u/s player down to about 185 u/s. Friction also applies, but over a few milliseconds it removes significantly less than the initial speed clamp.

A (true) perfect bhop is a jump that happens before WalkMove ever runs after landing. The player keeps all of their speed, and the only limit is the bunnyhop speed cap from `sv_enablebunnyhopping 0`, which cuts speed down to 286 u/s (1.1×260 , regardless of the weapon held). The player's height is also kept as is, which matters because the player can be up to 2 units above the ground when they land (more on that later).

If WalkMove runs even once before the jump, the player gets clamped down to max speed and snapped down to the floor.

3 How it worked in CS:GO

3.1 The jump check

Every tick, the game checks the jump button *before* running WalkMove.

```
// gamemovement.cpp, FullWalkMove
if (mv->m_nButtons & IN_JUMP)
    CheckJumpButton();
...
if (player->GetGroundEntity() != NULL)
    WalkMove();
```

Whether the player is on the ground is typically decided at the end of the previous tick. So if the player lands at some point during tick 100, tick 101 is the first tick where the game sees them on the ground, and it checks the jump button before WalkMove runs.

There are two things inside CheckJumpButton that can stop the jump. If the player is in the air, nothing happens and the press is not remembered. If jump was already held in the previous tick's input, the jump is blocked.

3.2 The perf window

Putting that together, a perfect bhop in CS:GO needs the jump input to be in the input packet of the first tick on the ground, and *not* in the input packet of the tick before that.

The perf window is therefore exactly one tick, which is 15.625ms on 64 tick and 7.8125ms on 128 tick. It doesn't matter where in the tick the player pressed jump or where in the tick they actually landed, only which tick's packet the press ended up in.

3.3 The jump cooldown

Because of the second rule, after a jump press the player needs a full tick with no jump input before the next press counts. How long they actually have to wait depends on where in the tick the first press was:

- If they pressed at the very start of a tick, they have to wait up to two ticks.
- If they pressed at the very end of a tick, the next tick can be empty even if they wait just over one tick.

On 128 tick, this means that a press that happens less than 7.8ms after the last press is always ignored, and the chance of it being ignored then drops linearly to 0% at 15.6ms.

See Eq. (A.2) for the probability that the next press counts.

3.4 Ideal scroll speed and perf rate

Scrolling puts a jump press roughly every X ms. For a scroll bhop to be perfect, one of those presses needs to fall into the one tick packet that matters, and the packet before it needs to be empty.

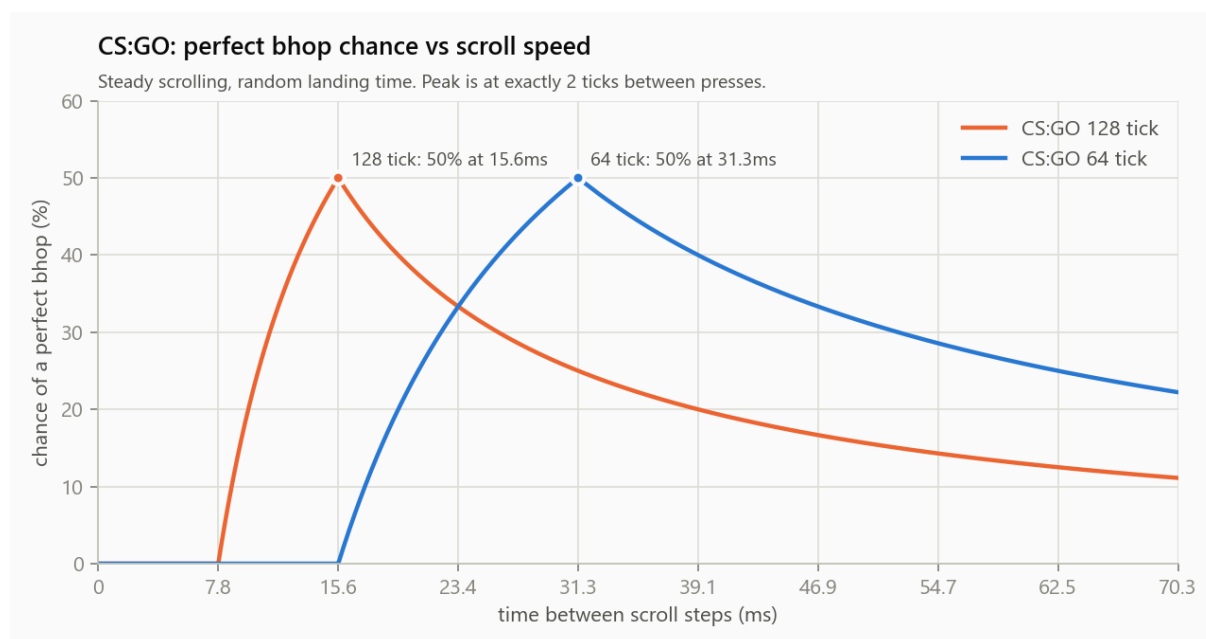


Figure 1: CS:GO: perfect bhop chance versus scroll speed.

Therefore, the best spacing between scroll steps is exactly 2 ticks (31.25ms on 64 tick, 15.625ms on 128 tick). That alternates between a packet with jump and a packet without, so every other tick is a valid perfect bhop tick, which gives a **50%** chance of hitting a perfect bhop.

Going either faster or slower reduces the odds. Slower means fewer presses land in the right tick (3 tick spacing gives 33%), and faster means some presses land in the tick right before the right one and block it (1.5 tick spacing also gives 33%).

The piecewise chance function is given by Eq. (A.1); its derivation is in Section A.2.1.

4 What changed in CS2

The subtick system in CS2 significantly complicates the situation.

4.1 Subtick basics

In CS2, inputs are timestamped with when they happened inside a tick (with a minimum resolution of $\sim 0.244\text{ms}$, or $1/4096\text{ s}$). Instead of simulating the whole tick in one go, the server splits the tick at every input and simulates each piece separately. These pieces are called subtick moves. For example, if the player presses A at 30% into the tick and jump at 80% into the tick, the tick is simulated as three subtick moves, 0-30%, 30-80% and 80-100%.

Each subtick move runs the full movement code. It checks the jump button at the start, then runs `WalkMove` (or air movement), then checks whether the player is on the ground at the end.

The tick gets split whenever the player presses or releases forward, back, left, right, jump or duck, and on attack inputs if the player's view moved. The game also adds a few splits on its own (landing related splits, friction and grenade related splits).

4.2 The bhop window

CS2 no longer uses the CS:GO “one tick on the ground” rule. Instead it has `sv_bhop_time_window` (default 0.0078125s , which is exactly one 128 tick). A jump pressed within half of that window (3.9ms) before or after landing is supposed to count as a bhop¹.

If the press is in the window and the player landed faster than their max speed, the game restores the horizontal speed the player had at the moment of landing, undoing the clamping that happened in between.

This means a jump inside the window can end up in several different ways, and only the first two are real perfect bhops.

¹This doesn't actually work as intended for multiple reasons (see the spaghetti sections).

1. A **frame perfect bhop** happens when the jump press is the input that ends the subtick move in which the player lands. The press either happens while the player is already within the last 2 units above the floor (see below), or right after they touch the floor with no other input in between. The jump starts at the moment of the press, before `WalkMove` runs, and speed and height are kept as is. This is the CS:GO idea of “jump on the first moment on the ground”, except the “moment” is now a client frame instead of a tick.
2. A **buffered perfect bhop** happens when the jump press comes slightly before landing, in the same subtick move as the landing, with no other input in between. Nothing happens at the press yet since the player is still in the air. The player lands at the end of that move, and the remembered press fires at the start of the next move (usually the start of the next tick), still before `WalkMove`. Speed and height are kept, but the jump starts at the tick boundary instead of at the press. This is the “before landing” half of `sv_bhop_time_window` doing what it’s supposed to do, but as explained below, it doesn’t really work most of the time.
3. A **restored bhop** happens when `WalkMove` ran at least once before the jump (because a tick ended or another input happened in between), but the player landed above max speed. Horizontal speed is restored from the landing speed, but the jump itself is different from a perfect one. The player gets snapped down to the floor by `WalkMove` before jumping, so if they were hovering (up to 2 units above the floor), that height is lost. The jump also starts later than the landing, and if the player was already on the floor (collision landing), that later start can instead give an ever so slightly higher jump than a perfect bhop would have.
4. A **deferred jump** happens when the player landed at or below max speed. The game refuses to jump until one full tick after landing, no matter when inside the window they pressed, so the player always spends a whole tick on the ground, just like a missed bhop in CS:GO. The pseudocode below shows why.

```
// OnJump
if ( bInWindow )
    bUseLandingVelocity = landingSpeed > maxSpeed;    // only restore above max speed

if ( bUseLandingVelocity )
{
    // restore landing speed
}
else
{
    // A regular jump can't happen in the same tick we landed
    if ( now < landed + 1 tick )
        return;
}
```

So if the player lands below max speed, a perfect bhop (either kind) is not possible at all, no matter how good their timing is. The two exceptions are ledgegrabs and jumpbugs, which skip this system entirely (and frame perfect rules apply instead).

Keep in mind that `maxSpeed` here includes the landing penalty (the modern jump code’s replacement for stamina), which temporarily lowers the max running speed right after a landing (to roughly 75% of normal after a flat ground bhop) and the speed of the next jump (to roughly 88%).

4.3 Bhop window spaghetti #1: The landing time is wrong

The bhop window is measured around the landing time. To get it, the game doesn’t just use the tick the player landed in. When it detects the player on the ground at the end of a subtick move, it solves the player’s fall motion to work out exactly when during that move they reached their final position. If the player hit the floor during the move, that’s the exact moment of contact, which is what the window is meant to be centered on.

The problem is that the game also puts the player on the ground when they haven’t hit the floor yet. At the end of each subtick move, it checks for ground up to **2 units below** the player, and if there is any, the player is considered on the ground even though they’re still hovering above it. The player’s final position for that move is in the air, so the calculation just returns the end of the move, and that becomes the landing time. **It’s earlier than the moment the player would have touched the floor.**

How much earlier that is depends on how fast the player is falling.
The time to fall the last 2 units is given by Eq. (A.3).

Table 1: Time to fall the last 2 units.

Fall speed	Time to fall the last 2 units
200 u/s	10.2ms
260 u/s	7.8ms
300 u/s	6.7ms
400 u/s	5.0ms
512 u/s	3.9ms

This gives a weird result. Since a jump press is itself a split, pressing jump while the player is inside those last 2 units will ground the player at the moment of the press, and record the landing as the press time. The press is then exactly on the landing time, so it's always inside the window.

In other words, **every press inside the last 2 units of the fall counts**, no matter how long that takes. At 260 u/s that is 7.8ms, which is almost twice the 3.9ms the window is supposed to give before landing.

If the player falls slower than 134 u/s, the last 2 units take longer than a whole tick. See Section A.3.1 for why it's 134 and not 128. A tick always ends while they're inside that range, so the recorded landing time is never the moment of contact. It's always the first tick end (or input) after the player gets within 2 units, up to a full tick or more before they would have touched the floor. This mostly happens when landing on a higher block.

However, there's a flip side. If a tick ends, or the player presses any other key, while they're inside those last 2 units, the player gets grounded at that moment instead. WalkMove then runs in the next move, and any jump after that is at best a restored bhop (no bonus height).

Landing while still going up (onto a ledge, or right at the top of a jump) is a separate case. The game only records a landing if the player was falling when the subtick move started. If they're still rising, no landing is recorded at all and the game keeps using the landing from before the jump, so the bhop window can never be hit. The player can still keep their speed by pressing jump before WalkMove runs, but there is no restore, no penalty and no window, only "press before the next input or tick end".

4.4 Bhop window spaghetti #2: Flawed jump buffering

The window is supposed to allow jumps up to 3.9ms before landing. The game remembers a jump press made in the air so that it can be used when the player lands. However, the game re-checks that remembered press at the start of every subtick move.

```
// CheckJumpButton, when no new jump press happened
TickFraction_t usable = GetLastUsableJumpPress();
if ( usable >= GetBhopWindowStart() )
{
    if ( usable < GetBhopWindowEnd() )           { OnJump( mv ); return; }
    if ( usable < GetLastLanded() + 1 tick )     { OnJump( mv ); return; }
}
// forget the press
m_nLastUsableJumpPressTick = JUMP_PRESS_TICK_NONE;
```

The window is centered on GetLastLanded(). While the player is still in the air, that is the landing *before* the current jump, which was most of a second ago. The remembered press is compared against that old landing, looks way too late, and gets thrown away. A few milliseconds later the player lands, and there's no press left to use.

The only way an early press survives is if the player lands in the same subtick move as the press, meaning the same tick, with no other input in between. If the player lands 50% into a tick, a press up to 3.9ms before landing works as intended. If the player lands 10% into a tick, only presses in that first 10% of the tick (1.6ms) work, and anything in the previous tick is forgotten. And if any other input happens between the press and the landing, the press is forgotten as well.

Combined with the 2 unit landing issue, the actual window ends up being shifted early, and its size depends on how fast the player is falling, where in the tick they land, and what other inputs they make around the landing.

The press-time ranges are listed in Section A.3.2 and formalized in Eq. (A.4) and Eq. (A.5).

intended:	[-3.9ms landing +3.9ms]
actual (260 u/s, no other input):	[-7.8ms landing ... +3.9ms]

The earlier part (inside the last 2 units) gives perfect bhops. The later part is often a restored bhop, depending on whether a tick ends in between.

The buffered kind of perfect bhop (a press made just before the last 2 units) almost never happens at normal fall speeds. A buffered press has to be made *before* the player enters the last 2 units, since a press inside them grounds the player on the spot and is frame perfect instead. It also can't be *more than 3.9ms before* the recorded landing time.

- If the player touches the floor during a subtick move, the landing time is the moment of contact. At normal fall speeds that is more than 3.9ms after the player entered the last 2 units, so no press made before them can be in the window.
- If a tick ends while the player is inside the last 2 units, they get grounded at that tick end, and the landing time moves earlier to that tick end. A buffered press then works if it falls between (tick end - 3.9ms) and the moment the player entered the last 2 units. That range only exists if the tick end comes less than 3.9ms after the player entered them, and it's 3.9ms - (time from entering the last 2 units to the tick end) wide.

For example, at 260 u/s the player enters the last 2 units at 13.0ms, the tick ends at 15.625ms, and they would touch the floor at 20.8ms. The landing time becomes 15.625, and a press between 11.7ms and 13.0ms is a buffered perfect bhop, a 1.3ms range. Averaged over where in the tick the landing happens, that's about 0.5ms worth of window. It only becomes significant on hard landings (faster than ~500 u/s), where the last 2 units take less than 3.9ms, so part of the 3.9ms before contact is always outside them.

In testing, buffered perfects make up about 3% of landings up to 512 u/s, then 6%, 11% and 14% at 600, 800 and 1000 u/s, exactly as predicted.

4.5 Frame perfect instead of tick perfect

In CS:GO, the player had a whole tick to get their jump in. The game only looked at which tick the input landed in.

In CS2, a perfect bhop almost always needs the jump to be the **very next input** after the player becomes grounded. The only exception is the rare buffered case (2) described above. Any other input in between (strafing with A/D, releasing W, attacking while turning...) splits the tick, the move after that split runs WalkMove, and the bhop is no longer perfect. At best it becomes a restored bhop (speed kept, different jump height, usually lower), and at worst the player gets clamped.

Since inputs are sent per client frame, this makes perfect bhops frame perfect rather than tick perfect. Players who strafe a lot right before landing, and players with higher fps (more frames means more places where inputs can split the tick), are more likely to lose the perfect bhop to one of their own inputs.

The buffered kind of perfect bhop has the same problem in the other direction. Any input between the early jump press and the landing splits the tick, and the remembered press gets thrown away (see above).

4.6 Crouchbugs and jumpbugs

A crouchbug is when the player lands by unducking in the air. Unducking in the air moves the player's feet down by 9 units. If the feet end up within 2 units of the floor, the player is classified as "on the ground" right away. So if the player unducks while their (ducked) feet are 9 to 11 units above the floor, they land instantly.

In CS, this landing happens inside the duck code, before the jump check and before WalkMove. Because the player is already on the ground when WalkMove runs, WalkMove resets their fall speed to zero, and the game never records a landing. That means no fall damage and no landing penalty (stamina in CS:GO). In CS2, the bhop window is never involved either.

A jumpbug is a crouchbug combined with a perfect bhop. The player unducks and jumps at the same time, so they land and jump off again without WalkMove ever running. Since no landing is recorded, the “can’t jump in the same tick you landed” rule is checked against the old landing and always passes, so a jumpbug keeps the player’s speed even below max speed. It is the purest form of the frame perfect case.

In CS:GO, both were tick perfect. The unduck had to be in the input packet of the tick where the player’s feet were 9 to 11 units above the floor. The player’s position is only checked once per tick, so if they fall more than 2 units per tick, that tick might not exist. On 64 tick that’s any fall faster than 128 u/s, so it failed on a lot of normal jumps. On 128 tick it’s any fall faster than 256 u/s, which is fine for most flat jumps but still fails on big drops.

Whether that tick exists only depends on the jump itself (where it started, how high it went, and where it lands). Jumps start on tick boundaries and movement is simulated in whole ticks, so the same jump always passes through the same heights at every tick. This makes CS:GO deterministic, since a given jump either always allows a crouchbug/jumpbug or never does, and players could learn which ones work. For a jumpbug, the jump also had to be in the same tick’s packet as the unduck, but anywhere within the same 7.8ms (128 tick) or 15.6ms (64 tick) was fine, and it could be done with a single alias bind.

In CS2, subtick makes this exceedingly complicated.

The unduck is processed at its exact subtick time, so there is always a moment where the feet are in the 9 to 11 unit range. But the player has to unduck within that range, which is only about 2 units of falling (~7.8ms at 260 u/s). There is no longer a fixed answer to “is this jumpbug possible”. Every attempt is possible in theory and depends on timing instead (and luck, since you need a frame to happen when you are within that range, see the note about low fps below).

For a jumpbug, the jump press has to be processed in the **same subtick move** as the unduck, which means both inputs have to be in the same client frame. If jump is pressed a frame later, the subtick move between the two inputs runs WalkMove on the ground and the player loses speed (commonly known as nopperf in CS:GO). If jump is pressed a frame earlier, the player is still in the air. The press is remembered, but then thrown away by the same stale landing check that breaks the pre-window (the player never has a new landing recorded here, so it’s compared against an old one from long ago).

This is therefore fps dependent. At 100 fps a frame is 10ms, at 300 fps it’s 3.3ms, and at 500 fps it’s 2ms. The lower the fps, the more time there is to get both inputs into the same frame, so frame perfect inputs get easier at low fps and harder at high fps. The same applies to the frame perfect kind of bhop, since fewer frames means fewer extra inputs that can split the tick near the landing.

Since inputs can only happen once per frame, if a frame is longer than the time the feet spend in the 9 to 11 unit range, there may be no frame at all where the unduck lands in range. This is the same problem CS:GO had with its ticks, except frames aren’t aligned to anything, so it’s random per attempt instead of fixed per jump.

CS2 blocks binding multiple commands to one key, so the CS:GO approach of putting both inputs in one alias doesn’t work. Pressing jump and releasing duck on two separate keys within the same frame, while also timing that frame to land inside a perf window, is exceedingly difficult to do consistently.

You can actually cheese this using the mouse wheel. Scroll inputs run both the + and - half of a bound +alias on the same scroll step, which is the same frame. So a +alias that does one input and a matching -alias that does the other, bound to the mouse wheel, sends both in the same frame on every scroll step. For example, with duck held on another key:

```
alias +jlb "+jump"
alias -jlb "-duck"
bind mwheeldown +jlb
// (don't actually use this, you will block yourself from jumping later because it needs a
-jump somewhere as well)
```

This turns the jumpbug from “two keys in the same frame” into “one scroll step at the right time”, with the right time still being the window where the feet are 9 to 11 units above the floor.

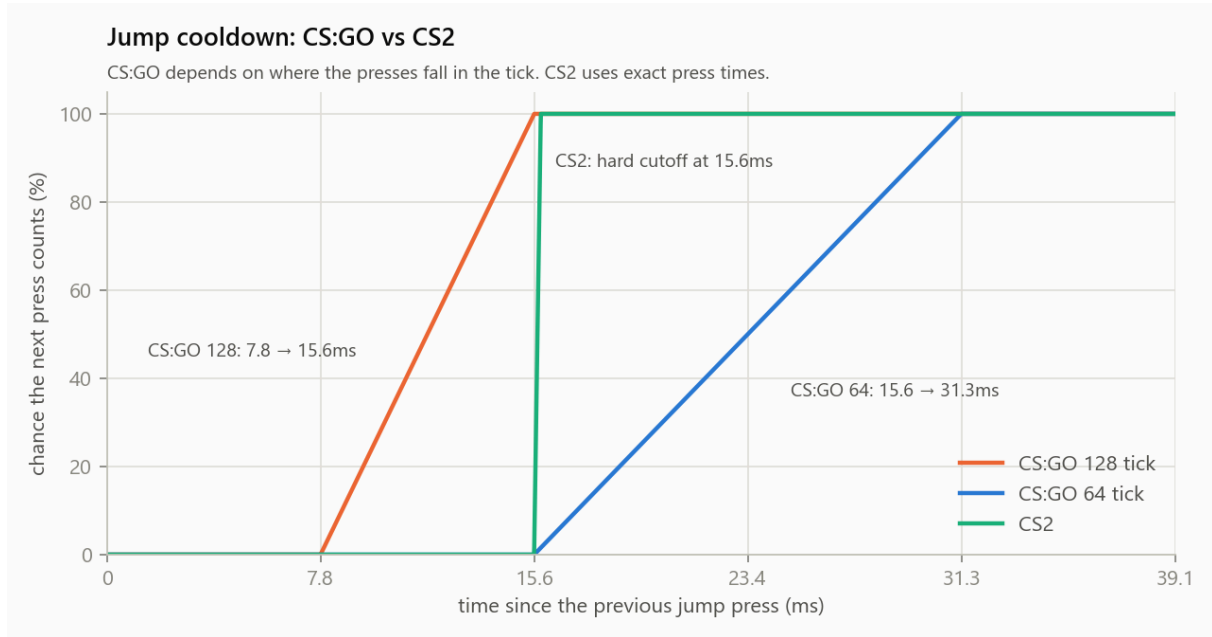


Figure 2: Jump cooldown: CS:GO and CS2.

4.7 The jump cooldown is... different

`sv_jump_spam_penalty_time` (15.625ms) ignores any jump press that comes within 15.625ms of the previous one. An ignored press still counts as “the previous press” for the next one, so scrolling too fast keeps resetting the timer and nothing gets through.

The check uses the subtick time of both presses, but the server rounds every subtick time to 1/64 of a tick (0.24ms) first. A gap of 15.625ms or less never counts, and a gap of more than about 15.9ms always counts. Anything in between is a lottery. Depending on how the two presses get rounded, the gap comes out as exactly one tick (ignored) or one tick plus 1/64 (counts). A steady scroll at 15.7ms only gets about 1 press in 3 through, and in testing 37% of landings got no jump at all.

See Eq. (A.15) for the rounding-dependent chance that a press counts.

This replaces the CS:GO ramp. On 128 tick CS:GO, gaps between 7.8ms and 15.6ms worked some of the time. In CS2 they never work. The CS2 cutoff sits exactly where CS:GO 128 tick reached 100%. **If you feel like you have to scroll differently to bhop in CS2, this is most likely why.**

4.8 Ideal scroll speed and perf rate in CS2

With CS:GO’s rule gone, scroll timing works differently. If the player scrolls with a steady gap between notches and the window is W ms wide, the chance that one of the presses lands inside the window is W divided by the gap, capped at 100%.

$$\text{chance} = \min\left(1, \frac{\text{window size}}{\text{gap between presses}}\right) \quad (\text{at most } 100\%) \quad (1)$$

Think of dropping a stick of length W onto a fence with posts every “gap” ms. The chance it touches a post is W / gap , or 100% if the stick is longer than the gap.

The gap can’t be shorter than the cooldown, and in practice has to be about 15.9ms because of the rounding lottery described above. So the ideal scroll speed is **about 16ms per notch** (about 60 notches per second). This is twice as fast as 64 tick CS:GO. A player with a 64 tick CS:GO scroll rhythm (31.25ms) only gets half the perf rate they could get.

At a 16ms gap, the average odds (assuming no other inputs and a random landing position within the tick) are shown below.

The formulas for each fall-speed range are in Section A.3.3; the conversion from windows to odds is given by Eq. (A.10) and Eq. (A.13).

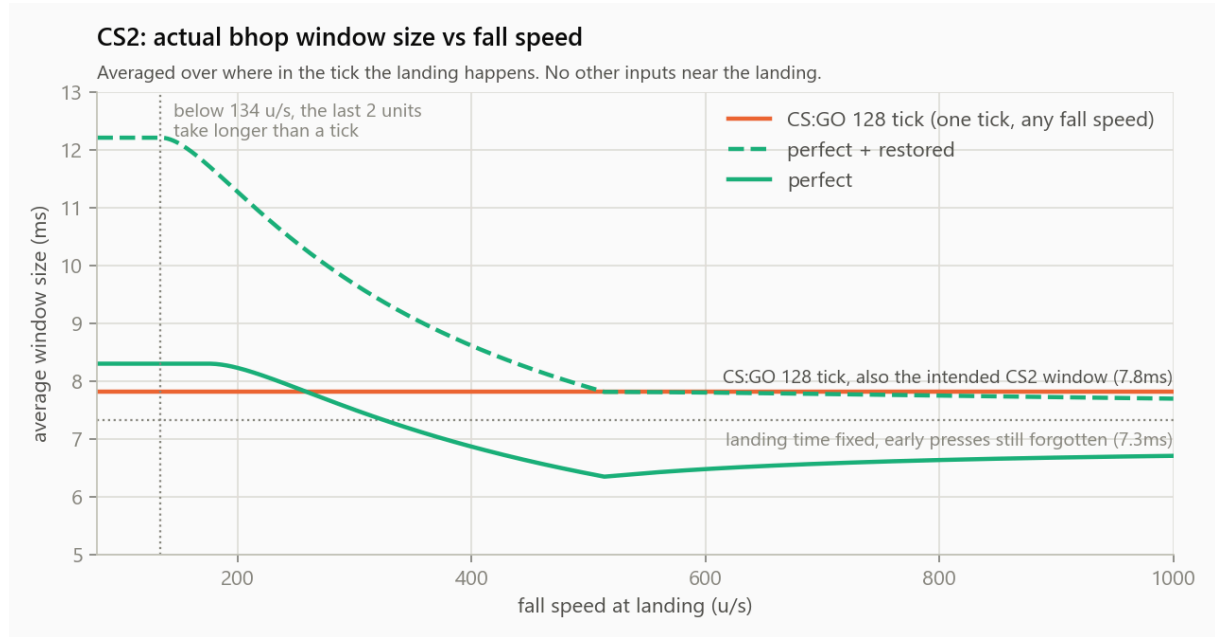


Figure 3: CS2: bhop window versus fall speed.

Table 2: Average bhop windows and odds at a 16ms gap between presses.

Fall speed	Perfect window	Perfect + restored window	Perfect odds	Perfect + restored odds	Measured (perfect / + restored)
160 u/s or slower	~8.3ms	~12.0ms	~52%	~74%	52–54% / 73–74%
200 u/s	~8.2ms	~11.3ms	~51%	~70%	52% / 70%
256 u/s ($\approx$ sustained bhop)	~7.8ms	~10.3ms	~49%	~64%	49% / 63%
300 u/s ($\approx$ first bhop)	~7.5ms	~9.7ms	~47%	~60%	48% / 60%
400 u/s	~6.9ms	~8.6ms	~43%	~54%	42% / 53%
512 u/s	~6.4ms	~7.8ms	~40%	~49%	41% / 49%
1000 u/s	~6.7ms	~7.7ms	~42%	~48%	43% / 48%
any speed, landing below max speed	0	0 (always deferred)	0%	0%	

Both graphs below include 128 tick CS:GO as a reference. Its window is always one tick (7.8ms) no matter how fast the player falls, which is also the size CS2’s window was meant to have, and at a 16ms scroll its perfect chance is 48.8% at any fall speed.

Against scroll speed, the CS2 perfect chance at 256 u/s ends up on almost the same curve as 128 tick CS:GO above 15.9ms, but it drops to zero below that instead of tapering off.

Almost all of the perfect window is the frame perfect kind. At 256 u/s it’s about 7.3ms frame perfect (~46%) and 0.5ms buffered (~3%).

For reference, if the landing time was calculated correctly (the moment the player touches the floor) but early presses were still forgotten, the window would be about 7.3ms (~46%) regardless of fall speed. With both issues fixed it would be the full 7.8ms (~49%).

These reference cases are formalized in Section A.3.6 and Eq. (A.17).

For anyone curious, here is roughly how the numbers are worked out. The full formulas behind every graph, and a step by step guide to deriving them, are in the appendix (Appendix A; see Section A.4).

Let d be the time to fall the last 2 units (under gravity, so slightly longer than $2 / \text{fall speed}$), and w be half the window (0.25 tick), with everything measured in ticks. If nothing splits the tick near the landing, the perfect + restored window is the last 2 units plus the 3.9ms after landing, $d + w$. If a tick ends inside the last 2 units, the player is grounded there instead, which cuts the window. Averaged over where in the tick the player lands, the cut is $\frac{d^2 - w^2}{2}$, so the average window is $d + w - \frac{d^2 - w^2}{2}$.

This is the normal-landing case of Eq. (A.7).

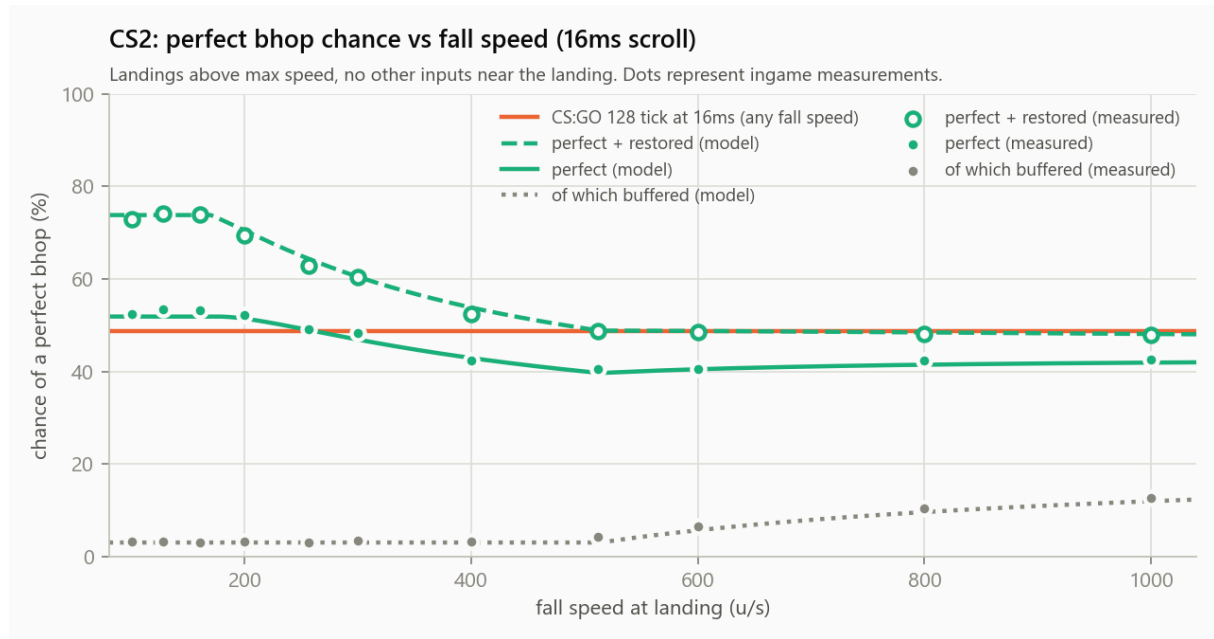


Figure 4: CS2: perfect bhop chance versus fall speed (16ms scroll).

The perfect part alone is everything before the first tick end (or other input) after the player enters the last 2 units, which averages $d + w - \frac{d^2}{2} - w \cdot d$. Of that, the buffered part only exists when a tick end falls less than w after the player enters the last 2 units, and averages $\frac{w^2}{2}$ (about 0.03 tick). The rest is frame perfect.

The corresponding averages are given by Eq. (A.7) and derived in Eq. (A.18).

The odds are $\frac{\text{window}}{\text{gap}}$ for each landing position, capped at 100%, then averaged. The cap matters for slow falls.

The cap is applied before averaging in Eq. (A.10); Eq. (A.12) gives the correction for windows longer than the press gap. When the window is longer than the gap between presses, one press is guaranteed to land in it, so averaging the window first and then dividing overstates the odds (by about 2.5 points at 128 u/s). Every other input near the landing shrinks the perfect part further and turns it into restored (or lost) bhops.

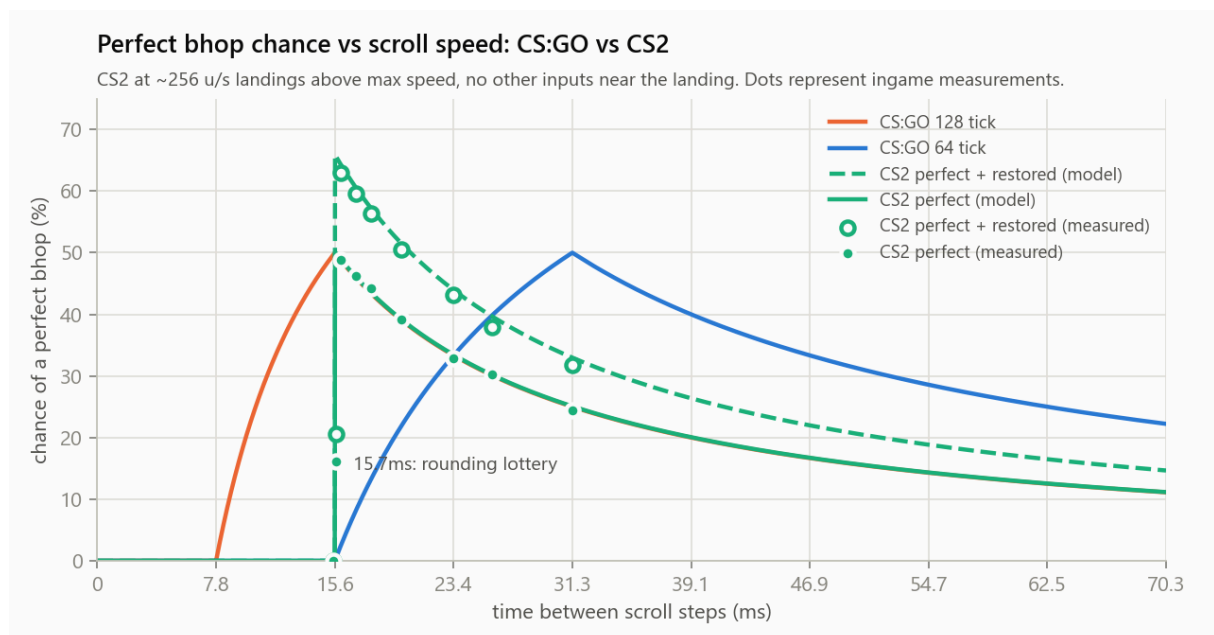


Figure 5: Perfect bhop chance versus scroll speed: CS:GO and CS2.

Unlike CS:GO, the odds are different for every jump depending on how fast the player falls, where in the tick they land, which other keys they press around the landing, and whether they landed above max speed. That is why perfect bhops in CS2 feel random even with consistent scrolling.

4.9 fps_max and consistent bhop timing

Inputs are timestamped by the client frame they're processed in, so with a frame cap every scroll step lands on a grid of $1000 / \text{fps_max}$ ms. However even if you scroll, the gap between two jumps always comes out as a whole number of frames. That makes the choice of fps_max matter, because that gap then has to clear the cooldown *after* the $1/64$ tick rounding described above.

A gap of g (in $1/64$ tick units, 0.244ms each) always rounds to either the whole number below g or the one above it, and it only passes if the rounded result is at least 65 (15.87ms). If N frames is at least 15.87ms, even the lower rounding passes, so every N -frame gap works and the rhythm is a steady N frames. If N frames is between 15.625ms and 15.87ms, some N -frame gaps round down and get ignored (resetting the timer) while others pass, so the best rhythm becomes " N frames, sometimes $N+1$ ", which is the lottery again.

On paper, a cap of exactly $N \times 64$ makes N frames exactly 15.625ms, so every N -frame gap would fail and the rhythm would always be $N+1$ frames. In practice this doesn't happen, because fps_max is only a maximum and the real frame rate flickers slightly below it. Frames are a bit longer than $1000 / \text{fps_max}$, which pushes N frames up into the lottery range, so a cap of exactly $N \times 64$ behaves like the lottery too.

That also works in your favour on the good caps. Because the real frame rate only ever drops below the cap, frames can only get longer, and a gap that passes at the cap's exact frame time keeps passing.

Scrolling a frame too early is possible at any fps and always fails. The point is to make sure that when you scroll at the right pace, the frame count you land on always passes, and that it's as close to 16ms as possible. Then the same scrolling gives the same result every time, instead of sometimes getting rounded into the lockout.

Multiples of 62.5 (125, 187.5, 250, 312.5, 375, 500) make N frames exactly 16ms, which is the shortest fixed rhythm that always passes. Right above each of them is a bad range where N frames falls into the lottery. Because the real frame rate sits a little under the cap, that range reaches from about $N \times 63$ to slightly above $N \times 64$.

Table 3: Frame caps for a consistent 16ms jump-input rhythm.

Frames between jumps	Good (N frames = 16ms)	Bad (lottery)
2	125	126 to just above 128
3	187.5	189 to just above 192
4	250	252 to just above 256
5	312.5	315 to just above 320
6	375	378 to just above 384
8	500	504 to just above 512

For example, this is what happens around 500 fps.

Table 4: Jump-input gaps and expected perfect bhop rates around 500 fps.

fps_max	Fastest gap that always passes	Gap	Expected perfect (sustained flat bhop)
490	8 frames	16.3ms	~48%
500	8 frames	16.0ms	~49%
505–512	8 frames, sometimes 9	lottery	lower
520	9 frames (as long as the real frame rate stays above 512)	17.3ms	~45%

Caps slightly *below* a multiple of 62.5 are also fine. They keep the same fixed rhythm, just a little slower (at 245 fps, 4 frames is 16.3ms, about 48% instead of 49%). There's no need to go below the multiple for safety, since frame time flicker at the cap only makes the gap longer, never shorter.

The game doesn't allow `fps_max` below 64, so a cap where a single frame is 16ms (62.5 fps) isn't possible.

The cap doesn't scroll for you, and you still need to hit roughly the right pace, about 16–20ms per notch. What a good cap does is make the outcome of a given scroll pace consistent. On a good cap, a scroll that lands on the 16ms frame count always gets through. On a bad cap, the same scroll sometimes or always gets rounded into the lockout.

4.10 Measured in game

All of the above was checked on a real CS2 server with a server-side test plugin.

1. Bots start by standing on flat ground.
2. For each landing, a bot is teleported into the air and dropped so it hits the floor at a chosen fall speed, with 275 u/s horizontal speed (above the speed cap).
3. The plugin writes the bot's input for every tick itself, including the subtick timestamps, so jump presses happen at exact, known times.
4. Each scroll step is a press and release in the same input packet, like a real mouse wheel, and no other inputs are made around the landing. The landing position within the tick and the scroll phase are randomized for every landing.

The outcome (perfect, buffered, restored, jump a tick later, or no jump) is read from the server's own movement state, which includes when the player was grounded, the recorded landing time, whether `WalkMove` ran before the jump, and the speed before and after the jump.

To put each landing at a random point in the tick without changing the fall speed, every drop starts with a random downward speed of up to 12.5 u/s (one tick of gravity), and the drop height is adjusted so the bot still hits the floor at exactly the target speed. The time spent in the last 2 units is then the same for every landing, and the landing position within the tick is uniformly random, which is what a player dropping from a fixed height gets.

The experimental procedure is summarized in Algorithm 1.

Algorithm 1: In-game measurement procedure, summarized from the description in this section.

Input: test points (v, T, N) : target fall speed, press gap, and number of landings.
Output: measured outcome rates at each test point.

```

for each test point (v, T, N):
    clear outcome counters
    repeat N times:
        start the bot on flat ground
        choose a random downward speed in [0, 12.5] u/s
        adjust drop height to reach the floor at speed v
        teleport and drop with horizontal speed 275 u/s
        randomize scroll phase relative to the landing
        write each tick's input with exact subtick timestamps:
            emit a jump press and release every T
            place each press/release pair in the same packet
            make no other inputs around the landing
        read the outcome from the server's movement state
        update the corresponding outcome counters
    report each outcome count divided by N

```

Note: Buffered bhops are also counted as perfect. Outcomes are determined from grounding and landing times, whether `WalkMove` ran before the jump, and speeds before and after the jump.

The first test varied the scroll speed, with 45,000 landings at about 256 u/s (5,000 per point, accurate to about $\pm 1.4\%$).

Table 5: Measured and modelled bhop rates versus gap between presses (45,000 landings).

Gap between presses	Perfect (measured / model)	of which buffered	Perfect + restored (measured / model)
15.5ms	0.0% / 0%	0.0%	0.0% / 0%
15.7ms	16.1% / lottery	1.2%	20.5% / lottery
16ms	48.8% / 48.8%	3.2%	62.9% / 64.1%
17ms	46.2% / 45.9%	2.8%	59.6% / 60.3%
18ms	44.3% / 43.4%	2.5%	56.3% / 56.9%
20ms	39.1% / 39.1%	2.4%	50.6% / 51.3%
23.4ms	32.9% / 33.4%	2.3%	43.1% / 43.8%
26ms	30.2% / 30.0%	1.7%	37.9% / 39.4%
31.25ms	24.4% / 25.0%	1.7%	31.8% / 32.8%

The second test varied the fall speed at a 16ms gap between presses, with 88,000 landings (8,000 per point, accurate to about $\pm 1.1\%$).

Table 6: Measured and modelled bhop rates versus fall speed at a 16ms gap (88,000 landings).

Fall speed	Perfect (measured / model)	of which buffered (measured / model)	Perfect + restored (measured / model)
100 u/s	52.4% / 51.9%	3.2% / 3.1%	72.9% / 73.8%
128 u/s	53.5% / 51.9%	3.2% / 3.1%	74.1% / 73.8%
160 u/s	53.2% / 51.9%	3.0% / 3.1%	73.9% / 73.8%
200 u/s	52.2% / 51.4%	3.2% / 3.1%	69.5% / 70.4%
256 u/s ($\approx$ sustained bhop)	49.2% / 49.0%	3.0% / 3.1%	62.9% / 64.4%
300 u/s ($\approx$ first bhop)	48.3% / 46.9%	3.5% / 3.1%	60.4% / 60.5%
400 u/s	42.3% / 42.9%	3.3% / 3.1%	52.5% / 53.8%
512 u/s	40.5% / 39.7%	4.2% / 3.1%	48.7% / 48.9%
600 u/s	40.6% / 40.5%	6.6% / 5.7%	48.6% / 48.8%
800 u/s	42.3% / 41.5%	10.4% / 9.6%	48.2% / 48.4%
1000 u/s	42.6% / 41.9%	12.6% / 12.0%	47.9% / 48.1%

The measured perfect rates sit up to about 1.5 points above the model at slow falls, which is about what the 1/64 tick rounding of press times (not included in the model) adds.

The logs also confirmed a few things directly.

- A press inside the last 2 units always records the landing at the press time and is always perfect, with the player jumping from up to 2 units above the floor.
- The window after the recorded landing ends exactly 0.25 tick (3.9ms) later, at every fall speed.
- A `WalkMove` before the jump clamps the player to the (lowered) max speed straight away (e.g. 275 $\rightarrow$ 189 u/s), and a restored bhop puts it back.
- Presses between 0.25 tick and 1 tick after the landing always jump exactly one tick after landing.

Appendix A: Formulas behind the bhop graphs

This appendix lists the closed-form formulas behind every graph in the main document, and then walks through how to derive them from scratch.

A.1 Notation

All CS2 times are measured in **ticks** (1 tick = 15.625ms). Multiply by 15.625 to get milliseconds.

Table A.1: Notation and reference values.

Symbol	Meaning	Value
g	gravity	800 u/s ²
v	fall speed when reaching the floor	u/s
w	half of sv_bhop_time_window	0.25 tick (3.9ms)
d	time spent falling through the last 2 units above the floor	ticks, see below
a	time from entering the last 2 units to the next tick end	uniform between 0 and 1 tick
T	time between two jump presses (scroll gap)	ticks, e.g. 16ms = 1.024 ticks

a is what captures “where in the tick the landing happens”. For a player landing from a given height, every value of a between 0 and 1 is equally likely, so every average below is an average over a .

The cooldown subsections locally reuse g for the press gap in milliseconds; elsewhere g denotes gravity.

A.2 CS:GO formulas

A.2.1 Perfect bhop chance vs scroll speed

Let t be the tick length (15.625ms on 64 tick, 7.8125ms on 128 tick) and G the scroll gap in ms. With $T = \frac{G}{t}$:

$$\text{chance} = \begin{cases} 0 & T < 1 \\ 1 - \frac{1}{T} & 1 \leq T \leq 2 \\ \frac{1}{T} & T \geq 2 \end{cases} \quad (\text{A.1})$$

A perfect bhop needs a press in the input packet of the first tick on the ground (probability $\frac{1}{T}$ for $T \geq 1$), and no press in the packet before it (probability $\min(1, T - 1)$ given the first). Multiplying gives $1 - \frac{1}{T}$ below 2 ticks and $\frac{1}{T}$ above. The peak is 50% at $T = 2$.

For example, on 64 tick with a 24ms gap, $T = 24/15.625 = 1.536$, so the chance is $1 - 1/1.536 = 34.9\%$. With a 40ms gap, $T = 2.56$ and the chance is $1/2.56 = 39.1\%$. With exactly 31.25ms, $T = 2$ and the chance is 50%.

The CS:GO window is always exactly one tick, regardless of fall speed. On 128 tick that is 7.8125ms, the same size as CS2’s intended sv_bhop_time_window. At a 16ms gap on 128 tick, $T = 2.048$, so the chance is $\frac{1}{2.048} = 48.8\%$ at any fall speed.

A.2.2 Jump cooldown

The chance that a press g ms after the previous one counts:

$$\text{chance} = \text{clamp}\left(\frac{g}{t} - 1, 0, 1\right) \quad (\text{A.2})$$

The second press only counts if a whole input packet between the two presses has no press in it. That happens with probability $\frac{g}{t} - 1$ for gaps between one and two ticks.

For example, on 128 tick a press 10ms after the previous one counts with probability $10/7.8125 - 1 = 0.28$ (28%), and a press 12ms after counts with probability $12/7.8125 - 1 = 0.54$ (54%). Anything under 7.8ms counts 0% of the time, and anything over 15.6ms counts 100% of the time.

A.3 CS2 formulas

A.3.1 Time spent in the last 2 units

The player reaches 2 units above the floor with speed $\sqrt{v^2 - 2 \cdot g \cdot 2}$. Under constant gravity, the time from there to the floor is:

$$d = 64 \cdot \frac{v - \sqrt{v^2 - 4g}}{g} \quad (\text{in ticks}) \quad (\text{A.3})$$

This is slightly longer than $\frac{2}{v}$ because the player is slower at the start of those 2 units. For example, $d = 0.506$ tick (7.9ms) at 256 u/s and $d = 1.054$ ticks at 128 u/s. Setting $d = 1$ and solving for v gives $v = (4g + 12.5^2)/25 = 134.25$ u/s, the landing speed at which the last 2 units take exactly one tick. The $2/v$ shortcut would give 128 u/s.

Plugging in 256 u/s: $\sqrt{256^2 - 3200} = \sqrt{62336} = 249.67$, so $d = 64 \cdot (256 - 249.67)/800 = 0.506$ tick, or 7.9ms. With the $2/v$ shortcut it would be $2/256$ s = 0.5 tick (7.8ms).

A.3.2 Window size for one landing

For a single landing with a given a , these are the lengths of the ranges of press times that give each outcome. They assume no other inputs near the landing and a landing above max speed.

Case A: a tick ends inside the last 2 units ($a < d$). The player is grounded while hovering at that tick end, and the landing time becomes that tick end.

Table A.2: Press-time ranges when a tick ends inside the last 2 units (Case A).

Outcome	Presses between	Length
Frame perfect	entering the last 2 units $\rightarrow$ tick end	a
Buffered perfect	(tick end - w) $\rightarrow$ entering the last 2 units	$\max(0, w - a)$
Restored	tick end $\rightarrow$ tick end + w	w

Case B: the player touches the floor before the next tick end ($a \geq d$). The landing time is the moment of contact.

Table A.3: Press-time ranges when contact precedes the next tick end (Case B).

Outcome	Presses between	Length
Frame perfect (in the last 2 units)	entering the last 2 units $\rightarrow$ contact	d
Frame perfect (after contact)	contact $\rightarrow$ min(tick end, contact + w)	$\min(w, a - d)$
Buffered perfect	(contact - w) $\rightarrow$ entering the last 2 units, but not before the previous tick end	$\max(0, \min(w - d, 1 - a))$
Restored	tick end $\rightarrow$ contact + w , if the tick end comes first	$\max(0, d + w - a)$

Adding them up:

$$\text{perfect}(a) = \begin{cases} \max(a, w) & a < d \\ d + \min(w, a - d) + \max(0, \min(w - d, 1 - a)) & a \geq d \end{cases} \quad (\text{A.4})$$

$$\text{total}(a) = \begin{cases} \max(a, w) + w & a < d \\ d + w + \max(0, \min(w - d, 1 - a)) & a \geq d \end{cases} \quad (\text{A.5})$$

total is perfect + restored, i.e. every jump that keeps the player's speed.

For example, at 256 u/s ($d = 0.506$ tick), a landing with $a = 0.3$ is Case A. Its perfect window is $\max(0.3, 0.25) = 0.3$ tick (4.7ms, all frame perfect), its restored window is $w = 0.25$ tick, and its total is 0.55 tick (8.6ms). A landing with $a = 0.1$ is also Case A, and its perfect window is $\max(0.1, 0.25) = 0.25$ tick, made of 0.1 tick of frame perfect presses and 0.15 tick of buffered presses. A landing with $a = 0.8$ is Case B. Its perfect window is $0.506 + \min(0.25, 0.294) + 0 = 0.756$ tick (11.8ms), its restored window is $\max(0, 0.506 + 0.25 - 0.8) = 0$, and its total is also 0.756 tick.

A.3.3 Average window size vs fall speed

These are the curves in `cs2_window_vs_fall_speed.png` (Figure 3): the averages of the formulas above over a from 0 to 1.

Hard landings, $d < w$ (faster than about 514 u/s):

$$\begin{aligned} \text{perfect} &= d - d^2 + w - \frac{w^2}{2} + \frac{(w-d)^2}{2} + (w-d)(1-w) \\ \text{buffered} &= w \cdot d - \frac{d^2}{2} + \frac{(w-d)^2}{2} + (w-d)(1-w) \\ \text{total} &= 2w \cdot d + (1-d)(d+w) + \frac{(w-d)^2}{2} + (w-d)(1-w) \end{aligned} \quad (\text{A.6})$$

Normal landings, $w \leq d \leq 1-w$ (about 175 to 514 u/s):

$$\begin{aligned} \text{perfect} &= d + w - \frac{d^2}{2} - w \cdot d \\ \text{buffered} &= \frac{w^2}{2} \\ \text{total} &= d + w - \frac{d^2 - w^2}{2} \end{aligned} \quad (\text{A.7})$$

Soft landings, $1-w < d \leq 1$ (about 134 to 175 u/s):

$$\begin{aligned} \text{perfect} &= \frac{1+w^2}{2} = 0.531 \text{ tick} \quad (8.3\text{ms}) \\ \text{buffered} &= \frac{w^2}{2} \\ \text{total} &= d + w - \frac{d^2 - w^2}{2} \end{aligned} \quad (\text{A.8})$$

Very soft landings, $d > 1$ (slower than about 134 u/s):

$$\begin{aligned} \text{perfect} &= \frac{1+w^2}{2} = 0.531 \text{ tick} \quad (8.3\text{ms}) \\ \text{buffered} &= \frac{w^2}{2} = 0.031 \text{ tick} \quad (0.5\text{ms}) \\ \text{total} &= \frac{1+w^2}{2} + w = 0.781 \text{ tick} \quad (12.2\text{ms}) \end{aligned} \quad (\text{A.9})$$

Plugging in one fall speed from each range: at 1000 u/s ($d = 0.128$, hard), perfect = 0.429 tick (6.7ms), buffered = 0.123 tick (1.9ms), and total = 0.493 tick (7.7ms). At 256 u/s ($d = 0.506$, normal), perfect = $0.506 + 0.25 - 0.128 - 0.127 = 0.502$ tick (7.8ms) and total = $0.756 - (0.256 - 0.0625)/2 = 0.659$ tick (10.3ms). At 160 u/s ($d = 0.827$, soft), perfect = 0.531 tick (8.3ms) and total = $1.077 - (0.683 - 0.0625)/2 = 0.766$ tick (12.0ms). At 100 u/s ($d = 1.403$, very soft), the values are the constants above, 0.531 tick perfect and 0.781 tick total.

The perfect window stops growing at $d = 1 - w$ because from there on, the frame perfect range in both cases is always “entering the last 2 units $\rightarrow$ next tick end”, which is a long whatever d is.

A.3.4 Perfect bhop chance

These are the curves in `cs2_rate_vs_fall_speed.png` (Figure 4) and `scroll_rate_comparison.png` (Figure 5).

With a steady scroll every T ticks and a random scroll phase, a range of press times that is W long contains a press with probability $\min(1, \frac{W}{T})$. This has to be applied to each landing (each a) before averaging, not to the average window:

$$\text{chance} = \int_0^1 \min\left(1, \frac{\text{window}(a)}{T}\right) da \quad (\text{A.10})$$

Perfect chance. $\text{perfect}(a)$ is never longer than 1 tick, and T is always longer than 1 tick (because of the cooldown), so the cap never applies:

$$\begin{aligned}\text{perfect chance} &= \frac{\text{perfect}}{T} \\ \text{buffered chance} &= \frac{\text{buffered}}{T}\end{aligned}\tag{A.11}$$

Perfect + restored chance. $\text{total}(a)$ can be longer than T for soft landings, so the part above T has to be removed first:

$$\text{excess} = \begin{cases} \frac{1}{2}\max(0, \min(d, 1) + w - T)^2 + (1 - d)\max(0, d + w - T) & w \leq d < 1 \\ \frac{1}{2}\max(0, 1 + w - T)^2 & d \geq 1 \\ 0 & d < w \end{cases}\tag{A.12}$$

$$\text{perfect + restored chance} = \frac{\text{total} - \text{excess}}{T}\tag{A.13}$$

The first term is Case A with $a + w > T$. The second is Case B, where the window is $d + w$ long for every a between d and 1. Without the excess term the chance is overstated for slow falls, by about 2.5 points at 128 u/s with a 16ms scroll.

For example, with a 16ms scroll ($T = 1.024$): at 256 u/s the perfect chance is $0.502/1.024 = 49.0\%$. Its total window never goes past $d + w = 0.756$ tick, which is shorter than T , so there is no excess and the perfect + restored chance is $0.659/1.024 = 64.4\%$. At 160 u/s, $d + w - T = 0.053$, so $\text{excess} = 1/2 \cdot 0.053^2 + (1 - 0.827) \cdot 0.053 = 0.0014 + 0.0092 = 0.0105$, and the perfect + restored chance is $(0.766 - 0.0105)/1.024 = 73.8\%$. At 128 u/s or slower ($d \geq 1$), $\text{excess} = 1/2 \cdot (1.25 - 1.024)^2 = 0.0255$, and the chance is $(0.781 - 0.0255)/1.024 = 73.8\%$, where leaving out the excess would give 76.3%.

Against scroll speed (`scroll_rate_comparison.png` (Figure 5)), the same formulas are used with v fixed at 256 u/s and T varying:

$$\text{chance} = \begin{cases} 0 & \text{gap} \leq 15.625 \text{ ms} \quad (\text{cooldown}) \\ \text{lottery, see below} & 15.625 \text{ ms} < \text{gap} < 15.87 \text{ ms} \\ \text{formulas above} & \text{gap} \geq 15.87 \text{ ms} \end{cases}\tag{A.14}$$

A.3.5 Jump cooldown in CS2

A press is ignored if it comes 15.625ms or less after the previous press (ignored presses still reset the timer). Both press times are first rounded to 1/64 tick (0.244ms), so for a gap of g ms:

$$\text{chance the press counts} = \text{clamp}\left(\frac{g - 15.625}{0.244}, 0, 1\right)\tag{A.15}$$

A gap between 15.625ms and 15.87ms rounds to exactly one tick (ignored) or one tick plus 1/64 (counts), depending on where the presses fall on the 1/64 grid. The chance of rounding up is the fraction of the 0.244ms step the gap covers.

For example, a gap of 15.7ms counts with probability $(15.7 - 15.625)/0.244 = 0.31$, a gap of 15.8ms with probability 0.72, and anything from 15.87ms up always counts.

For a steady scroll in that range, the gaps round to 64 or 65 units in a repeating pattern, and only the presses after a 65 count. At 15.7ms that is about 31% of presses, which left 37% of landings without any counted press after entering the last 2 units.

A.3.6 Reference values

If the landing time were correct, but early presses were still forgotten: the landing would always be the moment of contact, and a press before contact would only count within the same tick. With c the position of the contact within its tick:

$$\text{window}(c) = \min(w, c) + w \quad (\text{before contact} + \text{after contact})\tag{A.16}$$

$$\text{average} = 2w - \frac{w^2}{2} = 0.469 \text{ tick} \quad (7.3\text{ms})\tag{A.17}$$

If both worked as intended: $2w = 0.5$ tick (7.8ms), the same as one 128 tick.

A.4 How to derive the CS2 formulas yourself

Find, for one landing, every range of press times and which outcome each range gives, then average over where in the tick the landing happens.

Step 1: pick the units and the random variable. Measure time in ticks. The only thing that changes between two landings from the same height is where the tick grid falls relative to the fall. The cleanest way to describe that is a , the time from entering the last 2 units to the next tick end, which is uniform between 0 and 1.

As a running example, take a landing at 256 u/s with $a = 0.2$: the next tick ends 0.2 tick after the player enters the last 2 units.

Step 2: get the fall timing from the physics. Under constant gravity g , a player that reaches the floor at v reached 2 units above it with speed $\sqrt{v^2 - 4g}$, so it took $d = \frac{v - \sqrt{v^2 - 4g}}{g}$ seconds to fall the last 2 units (multiply by 64 for ticks). With B the moment of entering the last 2 units, $C = B + d$ is contact and $B + a$ is the next tick end.

For the example, $d = 64 \cdot (256 - \sqrt{256^2 - 3200}) / 800 = 0.506$ tick. Measuring from $B = 0$, contact is at $C = 0.506$ and the next tick end is at 0.2.

Step 3: write down the rules from the code. Every rule used here comes from reading the movement code:

1. The tick is split into subtick moves at every input and at every tick end. Each subtick move checks the jump button at its start, moves the player, then checks for ground at its end.
2. At the end of a subtick move, the player is put on the ground if there is floor within 2 units below them.
3. The landing time is the moment of contact if the player hit the floor during the move, and the end of the move if they were grounded while hovering.
4. A jump counts as a bhop if the press is within w of the landing time, i.e. in $[\text{landing} - w, \text{landing} + w)$.
5. If the jump fires before any WalkMove after landing, it's a perfect bhop. If a WalkMove ran first, it's restored.
6. A press made while in the air is remembered, but thrown away at the start of the next subtick move unless it's within the window of the *current* last landing. While still in the air, that is the landing before the jump.

Step 4: find the grounding moment. The player is grounded at the end of the first subtick move that ends while they're within 2 units (rule 2) or that contains the contact (rule 3). With only tick ends and jump presses as boundaries, that is:

- the tick end, if it comes before contact ($a < d$), and no jump press came first
- the jump press itself, if the press is between entering the last 2 units and that tick end
- otherwise, the end of the subtick move that contains the contact

In the example, the tick end (0.2) comes before contact (0.506), so this is Case A. Without an earlier press, the player is grounded while hovering at 0.2, and the landing time is 0.2. A press between 0 and 0.2 would ground the player at the press instead.

Step 5: go through every possible press time and classify it. For a single press at time p :

- **p before the last 2 units:** the player is in the air, so the press is remembered (rule 6). It survives only if no other boundary comes before the grounding, i.e. the grounding happens at the end of the same subtick move. It then counts as buffered if p is at most w before the landing time. In Case A that gives the range from (tick end - w) to entering the last 2 units, $\max(0, w - a)$. In Case B, contact is d after entering the last 2 units, so it only exists if $d < w$, and it can't start before the previous tick end: $\max(0, \min(w - d, 1 - a))$.
- **p inside the last 2 units and before the first tick end in there:** the press ends the subtick move, the player is grounded at p , the landing time is p , and the jump fires in the next subtick move before any WalkMove. That's frame perfect. This range is a long in Case A and d long in Case B.

- ***p* after contact, before the tick end (Case B):** the subtick move ending at *p* contains the contact, so the landing time is the contact and the press comes before WalkMove. It's frame perfect if $p - C < w$, so this range is $\min(w, a - d)$.
- ***p* after a tick end that grounded the player:** WalkMove already ran in the subtick move from the tick end to *p*. If *p* is still within *w* of the landing time, it's restored.

In the example, going from the earliest press to the latest:

- $p = -0.03$ (just before the last 2 units): the grounding at 0.2 ends the same subtick move, and $0.2 - (-0.03) = 0.23 < 0.25$, so it is a buffered perfect bhop. Presses from -0.05 to 0 work this way, a range of $w - a = 0.05$ tick (0.8ms).
- $p = 0.1$ (inside the last 2 units): the player is grounded at 0.1, so it is frame perfect. Presses from 0 to 0.2 work this way, a range of $a = 0.2$ tick (3.1ms).
- $p = 0.3$ (after the tick end): WalkMove ran from 0.2 to 0.3, and $0.3 - 0.2 < 0.25$, so it is restored. Presses from 0.2 to 0.45 work this way, a range of $w = 0.25$ tick (3.9ms).
- $p = 0.6$: more than *w* after the landing time, so the jump only fires one tick after landing.

Step 6: average over *a*. Each length from step 5 is a simple function of *a*, so its average is an integral from 0 to 1 over *a*, split at the points where max and min switch ($a = w$, $a = d$, $a = d + w$, $a = 1 - (w - d)$). For example, in the normal range ($w \leq d \leq 1 - w$). In Case A, the frame perfect range *a* and the buffered range $\max(0, w - a)$ add up to $\max(a, w)$. In Case B, the buffered term $\max(0, \min(w - d, 1 - a))$ is zero because $d \geq w$, so only the frame perfect ranges remain:

$$\begin{aligned}
 \text{average perfect} &= \int_0^d \max(a, w) da + \int_d^1 (d + \min(w, a - d)) da \\
 &= \left[w^2 + \frac{d^2 - w^2}{2} \right] + \left[d(1 - d) + \frac{w^2}{2} + w(1 - d - w) \right] \quad (\text{A.18}) \\
 &= d + w - \frac{d^2}{2} - w \cdot d
 \end{aligned}$$

For the example, this one landing has a perfect window of $0.2 + 0.05 = 0.25$ tick and a total of 0.5 tick. Plugging $d = 0.506$ into the average gives $0.506 + 0.25 - 0.128 - 0.127 = 0.502$ tick perfect (7.8ms) over all landing positions at 256 u/s. The same integral done for the total window gives $d + w - (d^2 - w^2)/2 = 0.756 - (0.256 - 0.0625)/2 = 0.659$ tick (10.3ms) perfect + restored.

Step 7: turn window lengths into chances. A steady scroll with gap *T* and a random phase puts a press into a range of length *W* with probability $\min(1, \frac{W}{T})$. Apply this per landing, then average over *a*. If *W* never exceeds *T*, that's just average $\frac{W}{T}$. Otherwise, subtract the parts of the windows longer than *T* first (the excess term).

For the example with a 16ms scroll ($T = 1.024$), this landing gives a perfect bhop with probability $0.25/1.024 = 24.4\%$ and keeps speed with probability $0.5/1.024 = 48.8\%$. Averaged over all landing positions at 256 u/s, the chances are $0.502/1.024 = 49.0\%$ perfect and $0.659/1.024 = 64.4\%$ perfect + restored.

The averaging operation in Step 7 can be approximated by Algorithm 2.

Algorithm 2: Numerical averaging of a selected bhop probability.

Input: fall speed *v*, press gap *T* in ticks, sample count *N*, and selected outcome (perfect, buffered, or perfect + restored).

Output: estimated probability of the selected outcome.

```

d <- fall time over the last 2 units, measured in ticks
sum <- 0
for i <- 0 to N - 1:
  a <- (i + 0.5) / N
  W <- window length for the selected outcome at (a, d)
  sum <- sum + min(1, W / T)
return sum / N

```

Note: Use Eq. (A.3) for *d*, Eq. (A.4) for the perfect window, and Eq. (A.5) for the perfect + restored window; the buffered ranges are listed in Section A.3.2. All window lengths and *T* use ticks. This approximates Eq. (A.10) for uniformly distributed *a*; the cooldown restrictions in Section A.3.5 still apply.

Step 8: check it. A small simulation that places the presses and the tick grid explicitly and applies the rules from step 3 should give the same numbers as the formulas. The in-game measurements in the main document (about 130,000 landings) match within about 1.5 points. The remaining difference comes from the 1/64 tick rounding of press times, which the continuous formulas leave out: when one edge of a window sits exactly on the rounding grid and counts as inside, the window is effectively half a grid step (0.12ms) wider.

For the example, the in-game measurement at 256 u/s with a 16ms scroll was 49.2% perfect and 62.9% perfect + restored, against 49.0% and 64.4% from the formulas.